

Deep Recurrent Neural Network-Based Observer for Uncertain Nonlinear Systems^{*}

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Abstract: Recurrent neural networks (RNNs) have gained popularity in various applications like handwriting recognition, time series prediction, and system identification due to their ability to retain state information for later use through temporal operations present in the RNN architecture. The dynamic behavior of RNNs makes them well-suited for the identification of nonlinear systems. Several training methods for adaptive RNN observers have been previously developed, but those methods are restricted to either RNNs satisfying a linear-in-the-parameters assumption or RNNs with a single hidden layer. While deep neural networks have been shown to yield improved function approximation performance compared to neural networks with a single hidden-layer, their complex structure and nested nonlinearities make real-time weight adaptation a difficult problem. In this paper, an adaptive deep RNN observer is developed for identification of a class of nonlinear systems. The developed deep RNN architecture adjusts in real-time to system uncertainties through a Lyapunov-derived weight adaptation law, where a filtered estimation error is designed and implemented in the weight adaptation law to relax full-state feedback requirements. A Lyapunov-based stability analysis is performed to ensure asymptotic convergence of the estimation errors and stability of the adaptive deep RNN architecture.

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1. INTRODUCTION

Unlike feedforward neural networks (NNs) which are memoryless and static, recurrent neural networks (RNNs) can capture the temporal dynamic behavior of an unknown system. Thus RNNs have gained popularity in various applications like handwriting recognition, speech recognition, time series prediction, and system identification (Petnehazi, 2019; Alex Graves, 2005; Hansika Hewamalage, 2021). Specifically, RNNs contain connections between the nodes that retain hidden state information for later use through temporal operations present in the RNN architecture. As a result, RNNs are a dynamic mapping and are better suited for dynamical system identification than feedforward NNs (Funahashi and Nakamura, 1993).

Various results employ RNNs for the identification of nonlinear systems, where the RNN is formulated as an observer to estimate the hidden states and identify the dynamics of the system. Compared to traditional model-based observers, RNN-based observers relax model knowledge requirements and can therefore be implemented for estimation of an unknown system. Many proposed methods employ offline optimization techniques to train the weights based on some loss function (Hanson et al., 2020; Hanson and Raginsky, 2020; Funahashi and

Nakamura, 1993; Sontag, 1998; Nikolakopoulou et al., 2022). Although these training methods have been successfully used in empirical studies, they lack analytical results with stability guarantees. Such offline methods pose limitations since training typically requires large, sufficiently rich datasets and state-derivative information, which is restrictive. Moreover, the accuracy of the resulting function approximation is based solely on offline training and does not consider real-time data. Due to the lack of continual learning, offline trained models can quickly get rendered obsolete whereas online methods consider real-time data, involve a closed-loop implementation, and provide stability guarantees under appropriate weight adaptation laws.

In contrast to offline training techniques, results in Dinh et al. (2017, 2014); Kosmatopoulos et al. (1995); Kim et al. (1997); Kim and Lewis (1999); Hovakimyan et al. (2002); Abdollahi et al. (2006) focus on the design and analysis of adaptive RNN architectures that are based on Lyapunov methods. These methods are restricted to either RNNs satisfying a linear-in-the-parameters assumption or RNNs with a single hidden layer. Although NNs with a single hidden layer are capable of approximating general nonlinear functions, deep NNs (DNNs) provide improved performance (LeCun et al., 2015). DNNs have recently grown in popularity for use in feedback control design due to their ability to model uncertain, complex systems. Results in Joshi and Chowdhary (2019); Joshi et al. (2020); Sun et al. (2022); Le et al. (2022); Patil et al. (2022) have investigated adaptive DNN-based control schemes to compensate for nonlinearities in system dynamics. The use of DNNs is motivated by the fact that DNNs are universal

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approximators and can therefore be used to estimate parametric uncertainties that do not satisfy the linear-in-the-parameters assumption required in most adaptive control methods. Moreover, DNNs are exponentially more efficient than shallow NNs regarding the total number of neurons required to achieve the same accuracy in function approximation (Rolnick and Tegmark, 2018). Thus, motivation exists to develop an adaptive deep RNN architecture.

In this paper, an adaptive deep RNN-based observer is developed to estimate a class of nonlinear systems. Leveraging the dynamic structure of RNNs, the developed observer design uses a deep RNN of an arbitrary depth to estimate unknown system states. Despite the challenges posed by the nested nonlinearities of deep RNNs, a Lyapunov-derived adaptation law is developed for the weights of each layer. Thus, the developed deep RNN architecture is able to adapt to system uncertainties in real-time. Motivated to relax full-state feedback requirements, a filtered estimation error is designed and implemented in the weight adaptation law. A Lyapunov-based stability analysis is performed to guarantee asymptotic convergence of the estimation errors and stability of the adaptive deep RNN architecture.

Notation and Preliminaries The space of essentially bounded Lebesgue measurable functions is denoted by $\mathcal{L}_\infty$. The right-to-left matrix product operator is represented by $\hat{\prod}$, i.e., $\hat{\prod}_{p=a}^m A_p = A_m \dots A_{a+1} A_a$ and $\hat{\prod}_{p=a}^m A_p = I$ if $a > m$, where I denotes the identity matrix. The Kronecker product is denoted by $\otimes$. Function compositions are denoted using the symbol $\circ$, e.g., given suitable functions g and h , $(g \circ h)(x) = g(h(x))$. The Filippov set-valued map defined in Paden and Sastry (1987, Equation 2b) is denoted by $K[\cdot]$. Consider a Lebesgue measurable and locally essentially bounded function $h : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$. Then, the function $y : \mathcal{I} \rightarrow \mathbb{R}^n$ is called a *Filippov solution* of $\dot{y} = h(y, t)$ on the interval $\mathcal{I} \subseteq \mathbb{R}_{\geq 0}$ if y is absolutely continuous on $\mathcal{I}$ and $\dot{y} \stackrel{a.a.t.}{\in} K[h](y, t)$, where *a.a.t.* denotes almost all time. The vectorization operator is denoted by $\text{vec}(\cdot)$, i.e., given $A \triangleq [a_{i,j}] \in \mathbb{R}^{n \times m}$, $\text{vec}(A) \triangleq [a_{1,1}, \dots, a_{1,m}, \dots, a_{n,1}, \dots, a_{n,m}]^T$. The vectorization operator satisfies the following properties (Bernstein, 2009, Proposition 7.1.9). Given any $A \in \mathbb{R}^{p \times a}$, $B \in \mathbb{R}^{a \times r}$, and $C \in \mathbb{R}^{r \times s}$,

$$\text{vec}(ABC) = (C^T \otimes A) \text{vec}(B), \quad (1)$$

and consequently,

$$\frac{\partial}{\partial \text{vec}(B)} \text{vec}(ABC) = (C^T \otimes A). \quad (2)$$

2. SYSTEM DYNAMICS

2.1 Model Dynamics

Consider an N-th order nonlinear system given in MIMO Brunovsky form as (Brunovsky, 1970)

$$\begin{aligned} \dot{x}_1 &= x_2 \\ &\vdots \\ \dot{x}_{N-1} &= x_N \\ \dot{x}_N &= f(x, u), \end{aligned} \quad (3)$$

where $x \triangleq [x_1^T x_2^T \dots x_N^T]^T \in \mathbb{R}^{Nn}$ and $u \in \mathbb{R}^m$ denote the generalized state and control input of the system, respectively, and $f : \mathbb{R}^{Nn} \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ denotes an unknown continuous function. The following assumptions facilitate the subsequent observer development.

Assumption 1. Let $z \triangleq [x^T u^T]^T$. There is a known, compact set $\mathcal{Z} \subseteq \mathbb{R}^{Nn} \times \mathbb{R}^m$ such that $z \in \mathcal{Z}$.

Assumption 2. The unknown function f is continuously differentiable.

Assumption 3. The system in (3) is observable.

Assumption 4. The states $x_1, \dots, x_{N-1}$ are assumed to be known and measurable.

The states $x_1, \dots, x_{N-1}$ are assumed to be available from sensor feedback. However, the higher order state x_N is not used by the subsequent development. Thus, an observer is designed to estimate x_N .

2.2 Deep Recurrent Neural Network Model

To compensate for the uncertainties in (3), a deep RNN model of the uncertain dynamical system is introduced in this section. A continuous-time deep RNN denoted by $\Psi(y, b, W_0, W_1, W_2, \dots, W_k) \in \mathbb{R}^{L_{k+1}}$ can be modeled as (Razvan Pascanu, 2014)¹

$$\begin{aligned} \dot{h} &= \Psi(y, b, W_0, W_1, W_2, \dots, W_k) \\ &= -bh + W_k^T \phi_k \circ \dots \circ W_1^T \phi_1 \circ W_0^T y, \end{aligned} \quad (4)$$

where $y \triangleq [h^T z_a^T]^T \in \mathbb{R}^{L_{k+1}+s+1}$, $h : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{L_{k+1}}$ denotes the hidden state, $z_a \triangleq [z^T 1]^T : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{s+1}$ denotes the augmented input, $z : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^s$ denotes the input, and $b \in \mathbb{R}_{>0}$ denotes a user-selected time constant. The smooth activation functions² and weight matrices at the j^{th} layer are denoted by $\phi_j : \mathbb{R}^{L_j} \rightarrow \mathbb{R}^{L_j}$ and $W_j^T \in \mathbb{R}^{L_{j+1} \times L_j}$ for all $j \in \{1, \dots, k\}$ and $j \in \{0, \dots, k\}$, respectively, where $L_j \in \mathbb{R}_{>0}, \forall j \in \{0, \dots, k\}$ denotes the number of neurons in the j^{th} layer. For DNNs with multiple types of activation functions at each layer, ϕ_j may be modeled as $\phi_j \triangleq [\varsigma_{j,1}, \varsigma_{j,2}, \dots, \varsigma_{j,L_{j-1}}, 1]^T$, where $\varsigma_{j,i} : \mathbb{R} \rightarrow \mathbb{R}$ for all $j \in \{1, \dots, k\}$ and $i \in \{1, \dots, L_j\}$ denotes the activation function at the i^{th} node of the j^{th} layer. To incorporate a bias term, z_a and ϕ_j are augmented with 1 for all $j \in \{1, \dots, k\}$. Since the hidden state is an input to the first hidden-layer, $W_0^T \in \mathbb{R}^{L_1 \times (L_{k+1}+s+1)}$.

To facilitate the subsequent analysis and address the challenges posed by the nested nonlinear parameterizations in the inner-layer RNN weights, a recursive representation of the RNN structure is developed. Let $\Phi_j(y, W_0, W_1, \dots, W_j) \in \mathbb{R}^{L_{j+1}}$ denote the output of the j^{th} layer defined as $\Phi_j \triangleq W_j^T \phi_j \circ W_{j-1}^T \phi_{j-1} \circ \dots \circ W_1^T \phi_1 \circ W_0^T y$. Thus, $\Phi_j(y, W_0, W_1, \dots, W_j)$ can be represented recursively as

¹ Converting the deep RNN model in Razvan Pascanu (2014) to continuous-time yields the model in (3).

² While the deep RNN in (4) does not account for nonsmooth activation functions for simplicity of notation, the switched analysis in Patil et al. (2022) can be used with the developed method to incorporate nonsmooth activation functions into the RNN architecture.

$$\Phi_j = \begin{cases} W_j^T \phi_j(\Phi_{j-1}), & j = \{1, \dots, k\}, \\ W_0^T y, & j = 0. \end{cases} \quad (5)$$

From (5), the deep RNN in (4) can be represented as $\Psi(y, b, W_0, W_1, W_2, \dots, W_k) = -bh + \Phi_k$.

The dynamical system in (3) can then be modeled using the deep RNN architecture defined in (4). The universal function approximation property states that the function space of the deep RNN architecture defined in (4) is dense in $\mathcal{C}(\mathcal{Z})$, where $\mathcal{C}(\mathcal{Z})$ denotes the space of continuous functions over $\mathcal{Z}$ (Kidger and Lyons, 2020, Theorem 3.2). Therefore, for any prescribed $\bar{\varepsilon} \in \mathbb{R}_{>0}$, there exist ideal weight matrices $W_j^{*T} \in \mathbb{R}^{L_{j+1} \times L_j}$, for all $j \in \{0, \dots, k\}$ such that $\|f(z) - \Psi(z, b, W_0^*, W_1^*, W_2^*, \dots, W_k^*)\| \leq \bar{\varepsilon}$ for all $z \in \mathcal{Z}$. Using the deep RNN architecture in (4), the dynamics in (3) can be rewritten as

$$\begin{aligned} \dot{x}_1 &= x_2 \\ &\vdots \\ \dot{x}_{N-1} &= x_N \\ \dot{x}_N &= -bx_N + W_k^{*T} \phi_k \circ \dots \circ W_1^{*T} \phi_1 \circ W_0^{*T} z_a + \varepsilon, \end{aligned} \quad (6)$$

where $W_j^{*T} \in \mathbb{R}^{L_{j+1} \times L_j}$, for all $j \in \{0, \dots, k\}$ denote ideal weight matrices and $\varepsilon : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^n$ denotes the function approximation error such that $\sup_{z \in \mathcal{Z}} \|\varepsilon(z)\| \leq \bar{\varepsilon}$. It is assumed there exists a known constant $\bar{\theta} \in \mathbb{R}_{>0}$ such that the layer weights can be bounded as $\|\theta^*\| \leq \bar{\theta}$, where $\theta^* \triangleq [\text{vec}(W_0^*)^T \dots \text{vec}(W_k^*)^T]^T$ (Lewis et al., 1996b, Assumption 1).

3. OBSERVER DEVELOPMENT

The objective is to design a deep RNN-based observer to estimate the dynamical system in (3) and ensure the estimated states $\hat{x} \triangleq [\hat{x}_1^T \hat{x}_2^T \dots \hat{x}_N^T]^T \in \mathbb{R}^N$ asymptotically converge to the actual states of the system. To quantify this objective, auxiliary identification errors $\tilde{x}_1, \dots, \tilde{x}_{N-1} \in \mathbb{R}^n$ with available sensor measurements are defined as

$$\begin{aligned} \tilde{x}_1 &\triangleq x_1 - \hat{x}_1, \\ \tilde{x}_2 &\triangleq \dot{\hat{x}}_1 + \tilde{x}_1, \\ \tilde{x}_j &\triangleq \dot{\hat{x}}_{j-1} + \tilde{x}_{j-1} + \tilde{x}_{j-2}, \quad \forall j = 3, \dots, N-1. \end{aligned} \quad (7)$$

To facilitate the subsequent analysis, an expression for $\tilde{x}_j$ for all $j \in \{3, \dots, N-1\}$ can be derived as

$$\tilde{x}_j = \sum_{i=0}^{j-1} a_{ji} (x_{i+1} - \hat{x}_{i+1}), \quad (8)$$

where the constant coefficients $a_{ji} \in \mathbb{R}$ are defined as (Xian et al., 2004, Eq. 10)

$$\begin{aligned} a_{j0} &= B(j) = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^j - \left(\frac{1-\sqrt{5}}{2} \right)^j \right], \\ \forall j &\in \{2, \dots, N-1\}, \\ a_{ji} &= \sum_{k=1}^{j-1} B(j-k-i+1) a_{k+i-1, i-1}, \\ \forall i &\in \{1, 2, \dots, j-2\}, j \in \{3, 4, \dots, N-1\}, \\ a_{j, j-1} &= 1, \quad \forall j \in \{1, 2, \dots, N-1\}. \end{aligned} \quad (9)$$

To compensate for the lack of direct measurements of x_N , the filtered estimation error $r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ is defined as

$$r \triangleq \dot{\tilde{x}}_{N-1} + \alpha \tilde{x}_{N-1} + \eta, \quad (10)$$

where $\alpha \in \mathbb{R}_{>0}$ denotes a user-selected constant and $\eta : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ denotes the output of a dynamic filter. The dynamic filter is designed as (Dinh et al., 2014)

$$\begin{aligned} \dot{\eta} &\triangleq p - (\alpha + k_r) \tilde{x}_{N-1}, \\ \dot{p} &\triangleq -(k_r + 2\alpha) p - \tilde{x}_f + \left((\alpha + k_r)^2 + 1 \right) \tilde{x}_{N-1}, \\ \dot{\tilde{x}}_f &\triangleq p - \alpha \tilde{x}_f - (\alpha + k_r) \tilde{x}_{N-1}, \end{aligned} \quad (11)$$

where $k_r \in \mathbb{R}_{>0}$ denotes a user-selected constant, $p \in \mathbb{R}^n$ denotes an internal filter variable, and $\tilde{x}_f \in \mathbb{R}^n$ denotes the auxiliary output of the filter. The filter variables p and $\tilde{x}_f$ are initialized such that $p(0) = (\alpha + k_r) \tilde{x}_{N-1}(0)$ and $\tilde{x}_f(0) = 0$, respectively. The filter in (11) uses $\tilde{x}_{N-1}$ as an input and produces the two filter outputs $\tilde{x}_f$ and η . The internal filter variable p is used to generate the signal η without involving the unmeasurable derivative of the estimation error $\dot{\tilde{x}}_{N-1}$.

From (10) and (11), the dynamic filter can be related to the unmeasurable auxiliary error r as

$$r = \dot{e} + \alpha e, \quad (12)$$

where $e : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ is an auxiliary error defined as $e \triangleq \tilde{x}_{N-1} + \tilde{x}_f$. To facilitate the subsequent stability analysis, taking the time derivative of η and using (10) and (11) yields

$$\dot{\eta} \triangleq -(\alpha + k_r) r - \alpha \eta + \tilde{x}_{N-1} - \tilde{x}_f. \quad (13)$$

3.1 System Identification Using Deep RNNs

RNNs are a dynamic mapping and are therefore well-suited for system identification. The recurrent feedback loop in an RNN enables it to approximate dynamic systems with any arbitrary degree of accuracy (Polycarpou, 1996). DNNs have been shown to be exponentially more efficient when compared to NNs with a single hidden layer regarding the number of neurons required to achieve the same accuracy (Rolnick and Tegmark, 2018). This motivates a deep RNN-based observer design. A subsequently designed adaptive weight update law trains the deep RNN to estimate the system dynamics in real-time. The observer for the nonlinear plant in (3) is designed using the deep RNN architecture in (4) as

$$\begin{aligned} \dot{\hat{x}}_1 &= \hat{x}_2 \\ &\vdots \\ \dot{\hat{x}}_{N-1} &= \hat{x}_N \\ \dot{\hat{x}}_N &= -b\hat{x}_N + \hat{W}_k^T \phi_k \circ \dots \circ \hat{W}_1^T \phi_1 \circ \hat{W}_0^T \hat{z}_a \\ &\quad + k_s \text{sgn}(e) + \chi, \end{aligned} \quad (14)$$

where $\hat{z}_a$ denotes the augmented input defined as $\hat{z}_a \triangleq [\hat{x}^T u^T 1]^T \in \mathbb{R}^{n+m+1}$, $\hat{W}_j^T \in \mathbb{R}^{L_{j+1} \times L_j}$ denotes the weight estimates for all $j \in \{0, \dots, k\}$, and $\chi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ denotes an auxiliary term designed to provide robustness to function reconstruction errors and inject coupling terms to account for related terms in the stability analysis. Based on the subsequent stability analysis, the auxiliary term χ is defined as

$$\chi \triangleq -(\gamma(k_r + \alpha) + 2\alpha)\eta + (\gamma - \alpha^2)\tilde{x}_{N-1} + \Upsilon, \quad (15)$$

where $\gamma \in \mathbb{R}_{>0}$ denotes a user-selected constant and $\Upsilon : \mathbb{R}_{>0} \rightarrow \mathbb{R}^n$ is an auxiliary term defined as

$$\Upsilon \triangleq \tilde{x}_{N-1} - \tilde{x}_f + \sum_{i=0}^{N-3} a_{N-1,i} (x_{i+2} - \hat{x}_{i+2}). \quad (16)$$

Taking the time derivative of (10) and substituting in (8), (14) and (15) into the $\dot{r}$ dynamics yields

$$\begin{aligned} \dot{\tilde{x}}_1 &= \tilde{x}_2 - \tilde{x}_1, \\ \dot{\tilde{x}}_j &= \tilde{x}_{j+1} - \tilde{x}_j - \tilde{x}_{j-1}, \quad \forall j \in \{2, \dots, N-2\}, \\ \dot{\tilde{x}}_{N-1} &= r - \alpha \tilde{x}_{N-1} - \eta, \\ \dot{r} &= f_e + \Phi^* - \hat{\Phi} + \varepsilon - k_s \text{sgn}(e) - \chi \\ &\quad + \alpha (r - \alpha \tilde{x}_{N-1} - \eta) + \dot{\eta} - \tilde{x}_{N-1} + \tilde{x}_f + \Upsilon, \end{aligned} \quad (17)$$

where the shorthand notations $\Phi^*(\hat{z}, \theta^*) \in \mathbb{R}^{L_{k+1}}$ and $\hat{\Phi}(\hat{z}, \hat{\theta}) \in \mathbb{R}^{L_{k+1}}$ are defined as $\Phi^* \triangleq \Phi_k(\hat{z}, \theta^*)$ and $\hat{\Phi} \triangleq \Phi_k(\hat{z}, \hat{\theta})$, for all $j \in \{0, \dots, k\}$, respectively, where $\hat{\theta} \triangleq [\text{vec}(\hat{W}_0)^T, \dots, \text{vec}(\hat{W}_k)^T]^T$. Let $\tilde{z} \triangleq z - \hat{z}$, where $\hat{z} \triangleq [\hat{x}^T u^T]^T \in \mathbb{R}^{Nn+m}$. The error $f_e : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ is defined as $f_e \triangleq \Psi(z, b, \theta^*) - \Psi(\hat{z}, b, \theta^*) = -b\tilde{x}_N + \Phi_k(z, \theta^*) - \Phi^*$ and can be bounded as $\|f_e\| \leq \rho_1(\|\tilde{z}\|)\|\tilde{z}\|$, where $\rho_1 : \mathbb{R} \rightarrow \mathbb{R}$ denotes a strictly increasing function (Kamalapurkar et al., 2014).

3.2 Adaptive Weight Update Laws

In this section, a Lyapunov-derived adaptation law is developed for the weights of the RNN observer. For notational brevity, let $\Phi_j^*(\hat{z}, W_0^*, W_1^*, \dots, W_j^*) \in \mathbb{R}^{L_{j+1}}$ and $\hat{\Phi}_j(\hat{z}, \hat{W}_0, \hat{W}_1, \dots, \hat{W}_j) \in \mathbb{R}^{L_{j+1}}$ be defined as $\Phi_j^* \triangleq \Phi_j(\hat{z}, W_0^*, W_1^*, \dots, W_j^*)$ and $\hat{\Phi}_j \triangleq \Phi_j(\hat{z}, \hat{W}_0, \hat{W}_1, \dots, \hat{W}_j)$, respectively, where $\tilde{\Phi}_j \triangleq \Phi_j^* - \hat{\Phi}_j$, for all $j \in \{0, \dots, k\}$. Let the shorthand notation $\tilde{\theta}$ be defined as $\tilde{\theta} \triangleq \theta^* - \hat{\theta}$. Based on the subsequent stability analysis, the RNN weight adaptation law is designed as

$$\dot{\hat{\theta}} \triangleq \text{proj}(\Gamma \hat{\Phi}^T e), \quad (18)$$

where $\Gamma \in \mathbb{R}^{\sum_{j=0}^k L_j L_{j+1} \times \sum_{j=0}^k L_j L_{j+1}}$ denotes a user-selected positive-definite gain matrix, $\hat{\Phi}' \in \mathbb{R}^{(Nn+m) \times \sum_{j=0}^k L_j L_{j+1}}$ is a short-hand notation denoting the Jacobian $\hat{\Phi}' \triangleq \frac{\partial \hat{\Phi}}{\partial \hat{\theta}}$, and $\text{proj}(\cdot)$ is a projection operator used to guarantee boundedness of the weight estimates (Krstic et al., 1995, Appendix E, Eq. E.4). The Jacobian $\hat{\Phi}'$ can be represented as

$$\hat{\Phi}' \triangleq [\hat{\Phi}'_0, \dots, \hat{\Phi}'_k], \quad (19)$$

where the shorthand notation $\hat{\Phi}'_j$ is defined as $\hat{\Phi}'_j \triangleq \frac{\partial \hat{\Phi}_j}{\partial \hat{\theta}}$, for all $j \in \{0, \dots, k\}$. Using (5), the chain rule, and the properties of the vectorization operator in (1) and (2), the terms $\hat{\Phi}'_0$ and $\hat{\Phi}'_j$ can be expressed as

$$\begin{aligned} \hat{\Phi}'_0 &\triangleq \left(\prod_{l=1}^k \hat{W}_l^T \hat{\phi}'_l \right) (I_{L_1} \otimes \hat{z}_a^T), \\ \hat{\Phi}'_j &\triangleq \left(\prod_{l=j+1}^k \hat{W}_l^T \hat{\phi}'_l \right) (I_{L_{j+1}} \otimes \hat{\phi}_j^T), \end{aligned} \quad (20)$$

for all $j \in \{1, \dots, k\}$, where the shorthand notations ϕ_j^* , $\hat{\phi}_j$, and $\hat{\phi}'_j$ are defined as $\phi_j^* \triangleq \phi_j(\hat{z}, W_0^*, W_1^*, \dots, W_j^*)$, $\hat{\phi}_j \triangleq \phi_j(\hat{z}, \hat{W}_0, \hat{W}_1, \dots, \hat{W}_j)$, and $\hat{\phi}'_j \triangleq \hat{\phi}'_j(\hat{\Phi}_{j-1})$ for all $j \in \{1, \dots, k\}$, respectively. The deep RNN in (4) is nonlinear in terms of the weights. To address the challenges posed by the deep RNN architecture, many results implement first-order Taylor series approximation-based techniques (Lewis et al., 1996b; Ge et al., 2002; Patil et al., 2022; Dinh et al., 2017). For the deep RNN given by (4), a first-order Taylor series approximation-based error model is given by (Lewis et al., 1996a, Eq. 22)

$$\tilde{\Phi} = \hat{\Phi}' \tilde{\theta} + \mathcal{O}^2(\hat{z}, \tilde{\theta}), \quad (21)$$

where $\mathcal{O}^2(\hat{z}, \tilde{\theta})$ denotes the higher-order terms. To facilitate the subsequent stability analysis, let the shorthand notation $\tilde{\Phi}_*$ be defined as $\tilde{\Phi}_* \triangleq \Phi_j(z, \theta^*) - \Phi_j(z, \hat{\theta})$. Using first-order Taylor series approximation-based techniques yields

$$\tilde{\Phi}_* = \hat{\Phi}'_* \tilde{\theta} + \mathcal{O}^2(z, \tilde{\theta}), \quad (22)$$

where the shorthand notation $\hat{\Phi}'_*$ is defined as $\hat{\Phi}'_* \triangleq \frac{\partial \Phi_j(z, \theta^*)}{\partial \theta}$ and $\mathcal{O}^2(z, \tilde{\theta})$ denotes the higher order terms. Adding and subtracting $\mathcal{O}^2(z, \tilde{\theta})$ and substituting in (21), the $\dot{r}$ dynamics in (17) can be rewritten as

$$\begin{aligned} \dot{r} &= \tilde{N}_1 + \hat{\Phi}' \tilde{\theta} - k_s \text{sgn}(e) - \chi + \alpha (r - \alpha \tilde{x}_{N-1} - \eta) \\ &\quad + \dot{\eta} - \tilde{x}_{N-1} + \tilde{x}_f + \Upsilon + N_1, \end{aligned} \quad (23)$$

where the auxiliary terms $N_1 \in \mathbb{R}^n$ and $\tilde{N}_1 \in \mathbb{R}^n$ are defined as

$$\begin{aligned} N_1 &\triangleq \mathcal{O}^2(z, \tilde{\theta}) + \varepsilon, \\ \tilde{N}_1 &\triangleq f_e + \mathcal{O}^2(\hat{z}, \tilde{\theta}) - \mathcal{O}^2(z, \tilde{\theta}), \end{aligned} \quad (24)$$

respectively. The higher-order terms in (21) are defined in terms of $\tilde{z}$ and therefore cannot yet be proven bounded. Adding and subtracting $\mathcal{O}^2(z, \tilde{\theta})$ allows for the combined higher-order terms to be expressed in terms of the state $\tilde{z}$. Thus, using Assumption 1 and the projection operator, the nonlinear terms in the $\dot{r}$ dynamics are segregated into auxiliary terms N_1 , which can be bounded as $\|N_1\| \leq \xi_1$, and $\tilde{N}_1$, which has a state-dependent bound $\|\tilde{N}_1\| \leq \rho_2(\|\tilde{z}\|)\|\tilde{z}\|$ based on the inequality in (Kamalapurkar et al., 2014), where $\xi_1 \in \mathbb{R}_{>0}$ denotes a known constant and $\rho_2 : \mathbb{R} \rightarrow \mathbb{R}$ denotes a strictly increasing function.

4. STABILITY ANALYSIS

The state estimation errors are guaranteed to asymptotically converge in the following theorem.

To facilitate the subsequent stability analysis, let the auxiliary terms $N_2 \in \mathbb{R}^n$ and $\tilde{N}_2 \in \mathbb{R}^n$ be defined as $N_2 \triangleq \hat{\Phi}'_* \tilde{\theta}$ and $\tilde{N}_2 \triangleq \hat{\Phi}' \tilde{\theta} - \hat{\Phi}'_* \tilde{\theta}$, respectively. From Assumption 1, the projection operator, and (20)-(22), the auxiliary terms N_2 and $\tilde{N}_2$ can be bounded as $N_2 \leq \xi_2$ and $\|\tilde{N}_2\| \leq \rho_3(\|\tilde{z}\|)\|\tilde{z}\|$, respectively, and the time derivative of $N_1 + N_2$ can be bounded as $\dot{N}_1 + \dot{N}_2 \leq \xi_3 + \rho_4(\|\tilde{z}\|)\|\tilde{z}\|$, where $\xi_2, \xi_3 \in \mathbb{R}_{>0}$ denote known constants and $\rho_3, \rho_4 : \mathbb{R} \rightarrow \mathbb{R}$ denote strictly increasing functions.

Theorem 1. Let the auxiliary constants λ_1, λ_2 , and λ_3 be defined as $\lambda_1 \triangleq \frac{1}{2} \min\{1, \gamma, \alpha\}$, $\lambda_2 \triangleq \frac{1}{2} \max\{1, \gamma, \alpha\}$, and $\lambda_3 \triangleq \min\{\gamma, \alpha\gamma, k_r\}$, respectively. Consider the dynamical system in (3) with Assumptions 1-3. The observer in (14) and the adaptation law in (18) ensure asymptotic estimation error convergence in the sense that $\|x_j - \hat{x}_j\| \rightarrow 0$, as $t \rightarrow \infty$ for all $j \in \{1, \dots, N\}$ provided the following gain conditions are satisfied

$$\begin{aligned} \alpha &\geq \frac{1}{2}, \\ k_s &> \xi_1 + \xi_2 + \frac{1}{\alpha - \lambda_P} (\alpha \xi_2 + \xi_3), \\ \lambda_3 &> \rho \left(\sqrt{\frac{\lambda_2}{\lambda_1}} \|\zeta(t_0)\| \right), \end{aligned} \quad (25)$$

where $\rho : \mathbb{R} \rightarrow \mathbb{R}$ denotes a strictly increasing, invertible function such that $\rho(\|\tilde{z}\|) \geq \rho_2(\|\tilde{z}\|) + \rho_3(\|\tilde{z}\|) + \rho_4(\|\tilde{z}\|)$ and $\zeta \in \mathbb{R}^\psi$ is defined as $\zeta \triangleq \begin{bmatrix} \tilde{x}_1^T & \tilde{x}_2^T & \dots & \tilde{x}_{N-1}^T & r^T & \eta^T & \tilde{x}_f^T & P & \tilde{\theta}^T \end{bmatrix}^T$, where $\psi \triangleq Nn + 2n + 1 + \sum_{j=0}^k L_j L_{j+1}$.

Proof. Consider the candidate Lyapunov function $\mathcal{V}_L : \mathbb{R}^\psi \rightarrow \mathbb{R}_{\geq 0}$ defined as

$$\begin{aligned} \mathcal{V}_L(\zeta) &\triangleq \frac{\gamma}{2} \sum_{j=1}^{N-1} \tilde{x}_j^T \tilde{x}_j + \frac{1}{2} r^T r + \frac{\gamma}{2} \eta^T \eta \\ &\quad + \frac{\gamma}{2} \tilde{x}_f^T \tilde{x}_f + P + \frac{\alpha}{2} \tilde{\theta}^T \Gamma \tilde{\theta}, \end{aligned} \quad (26)$$

where $P : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is defined as

$$\begin{aligned} P(t) &= e^{-\lambda_P t} * \left((\alpha - \lambda_P) \left(k_s \|e\|_1 - e^T \left(\dot{N}_1 + \dot{N}_2 \right) \right) \right. \\ &\quad \left. - \alpha e^T N_2 + e^T \left(\dot{N}_1 + \dot{N}_2 \right) \right) + k_s \|e\|_1 \\ &\quad - e^T \left(\dot{N}_1 + \dot{N}_2 \right) + e^{-\lambda_P t} * \left(\rho_4(\|\tilde{z}\|) \|\tilde{z}\|^2 \right), \end{aligned} \quad (27)$$

where $\|\cdot\|_1$ denotes the 1-norm, $\lambda_P \in \mathbb{R}_{>0}$ denotes a constant, and $*$ denotes the convolution operator. The candidate Lyapunov function satisfies

$$\lambda_1 \|\zeta\|^2 \leq \mathcal{V}_L \leq \lambda_2 \|\zeta\|^2, \quad (28)$$

where λ_1 and λ_2 are defined in the theorem statement.

Using the developed closed-loop error system in (13) and (15)-(21) and provided the sufficient conditions in (25) are satisfied, the candidate Lyapunov function in (26) can be bounded as

$$\dot{\mathcal{V}}_L \leq -(\lambda_3 - \rho(\|\tilde{z}\|)) \|\tilde{z}\|^2. \quad (29)$$

The expression in (29) can be rewritten as

$$\dot{\mathcal{V}}_L \leq -W(\zeta) \leq -c \|\tilde{z}\|^2, \quad \forall \zeta \in \mathcal{D}, \quad (30)$$

with some constant $c \in \mathbb{R}_{>0}$, where $W : \mathbb{R}^\psi \rightarrow \mathbb{R}$ is a continuous positive semidefinite function and $\mathcal{D} \in \mathbb{R}^\psi$ denotes an open and connected set defined as

$$\mathcal{D} \triangleq \{\sigma \in \mathbb{R}^\psi \mid \|\sigma\| < \rho^{-1}(\lambda_3)\}.$$

Whenever $\zeta \in \mathcal{D}$, $\|\zeta\| < \rho^{-1}(\lambda_3)$ and therefore $\|\tilde{z}\| < \rho^{-1}(\lambda_3)$. From the definition of $\mathcal{D}$ and (26), if $\zeta \in \mathcal{D}$, $\lambda_3 > \rho \left(\sqrt{\frac{\lambda_2}{\lambda_1}} \|\zeta(t_0)\| \right)$. From (30), this implies that there exists a $c \in \mathbb{R}_{>0}$ that satisfies (30). Moreover, this implies larger values of λ_3 expand the size of $\mathcal{D}$. Since $\mathcal{V}_L$ is nonincreasing, $\|\zeta(t)\| \leq \sqrt{\frac{\mathcal{V}_L(t)}{\lambda_1}} \leq \sqrt{\frac{\mathcal{V}_L(t_0)}{\lambda_1}}$ and therefore $\sqrt{\frac{\mathcal{V}_L(t_0)}{\lambda_1}} \leq \rho^{-1}(\lambda_3)$ is sufficient to obtain $\zeta \in \mathcal{D}$.

Since $\mathcal{V}_L(t_0) \leq \lambda_2 \|\zeta(t_0)\|^2$, $\sqrt{\frac{\mathcal{V}_L(t_0)}{\lambda_1}} < \rho^{-1}(\lambda_3)$ implies $\sqrt{\frac{\lambda_2}{\lambda_1}} \|\zeta(t_0)\| < \rho^{-1}(\lambda_3)$ and $\|\zeta(t_0)\| < \sqrt{\frac{\lambda_1}{\lambda_2}} \rho^{-1}(\lambda_3)$. Therefore $\mathcal{S} \triangleq \{\sigma \in \mathcal{D} \mid \|\sigma\| \leq \sqrt{\frac{\lambda_1}{\lambda_2}} \rho^{-1}(\lambda_3)\}$ denotes the set where $\zeta(t_0)$ should lie to guarantee that $\zeta(t) \in \mathcal{D}$ for all $t \in [0, \infty)$. Using (26) and the fact that $\dot{\mathcal{V}}_L \leq 0$ implies $\tilde{x}_j, \tilde{x}_f, \eta, r, P, \tilde{\theta} \in \mathcal{L}_\infty$. Therefore, the observer $\hat{x} \in \mathcal{L}_\infty$.

By property of the projection operator, $\hat{\theta} \in \mathcal{L}_\infty$. Using (26), (28), and (30), the extension of LaSalle-Yoshizawa corollary in Fischer et al. (2013, Corollary 1) can be invoked to show $\|\tilde{x}_j\| \rightarrow 0$, $\|\tilde{x}_f\| \rightarrow 0$, $\|\eta\| \rightarrow 0$, and $\|r\| \rightarrow 0$ as $t \rightarrow \infty$ for all $j \in \{1, \dots, N-1\}$. From (8) and (10), it can be further shown that $\|x_N - \hat{x}_N\| \rightarrow 0$ as $t \rightarrow \infty$. Provided the gain conditions in (25) are satisfied according to the initial condition, the region of attraction can be made arbitrarily large to include any initial condition by increasing the gains γ, α , and k_r accordingly.

5. CONCLUSIONS

Leveraging the dynamics structure of RNNs, an adaptive deep RNN-based observer is developed for the estimation of a class of nonlinear systems. A Lyapunov-derived adaptation law is developed for the weights of each layer of the deep RNN observer using filtered estimation errors to compensate for unknown system states. Asymptotic convergence of the estimation errors and stability of the adaptive deep RNN architecture are guaranteed via a Lyapunov-based stability analysis.

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